MKALAMA DISTRICT COUNCIL
FORM ONE TERMINAL EXAM.
BASIC MATHEMATICS
MARKING SCHEME.

1. (a) Branches of mathematics (05 marks)
   - i) Algebra
   - ii) Geometry
   - iii) Arithmetic

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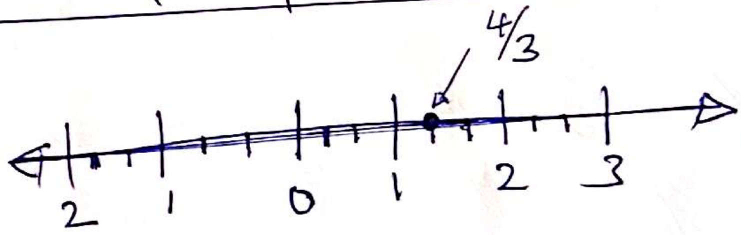
(b) Activities that involves use of mathematics concepts in Geography are

- (i) Measuring of distance between two points on a map.
- (ii) Measuring of height of mountains
- (iii) Calculation of surface area on a map (05 marks)

2. (a) (05marks)



(b) $1{\cdot}5$ or $1\frac{1}{2}$ or $\frac{3}{2}$ (05 marks)

3. (a) i) $a > c$ (2.5 marks)

ii) $a > c$ (2.5 marks)

3. ⓑ Given $\frac{1}{4}$ and $\frac{3}{7}$

LCM of 4 and 7 is 28.

$\frac{1}{4} \times 28 = 7$

$\frac{3}{7} \times 28 = 3 \times 4 = 12$

$\therefore \frac{3}{7}$ is greater than $\frac{1}{4}$ (0.5 marks)

4. ⓐ Irrational numbers are (1 mark each.)

$\frac{\sqrt{2}}{2}$, π, $\sqrt{3}$, $\frac{\pi}{4}$ and $\frac{-\sqrt{5}}{3}$ (05 marks)

ⓑ Equivalent rational numbers are

0.75, $6/8$, $\frac{9}{12}$, $\frac{12}{16}$, $\frac{15}{20}$, etc. (05 Marks)

(1 mark each)

5. ⑤ ⓐ i) 0.3

ii) 18

iii) 64000

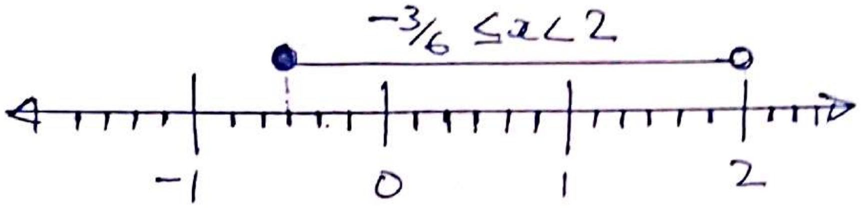
} 05 marks

ⓑ $0.3143 \times 6.06 = 1.904658$

$\simeq 1.905$

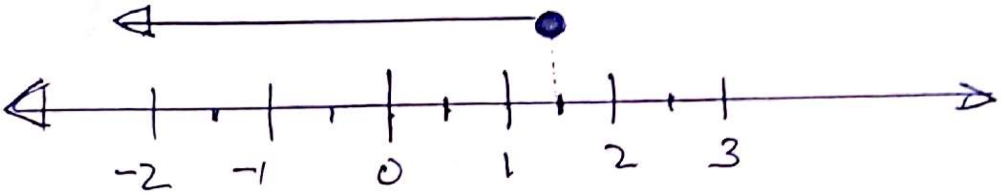
} 05 marks

6. (a) $-\frac{3}{6} \le x < 2$



(5 marks)

(b) $x \le \frac{3}{2}$



(5 marks)

7. (a) $2.\dot{3} + 0.\dot{3} - 2.\dot{4}$

$= (2.333\ldots + 0.333\ldots) - (2.444\ldots)$

$= (2.666\ldots) - (2.444\ldots)$

$= 0.222\ldots$

(02$\frac{1}{2}$ marks)

Let $x = 0.222\ldots \quad \ldots ⓘ$

$10x = 2.222\ldots \quad \ldots (ii)$

$10x - x = 2.222 - 0.222$

$\frac{9\cancel{x}}{9} = \frac{2}{9}$

$x = 2/9.$

02 marks

$\therefore 2.\dot{3} + 0.\dot{3} - 2.\dot{4} = \frac{2}{9}$

00$\frac{1}{2}$ marks

7 (b) i)

$$\begin{array}{r} 0.33\ldots \\ 3\overline{)\,1} \\ \underline{0} \\ 10 \\ \underline{-9} \\ 10 \\ \underline{9} \\ 1 \end{array}$$

$\therefore \frac{1}{3} = 0.\dot{3}$

$\Big\}$ $\left(02\frac{1}{2} \text{ marks}\right)$

ii) $1\frac{5}{9} = \frac{14}{9}$

$$\begin{array}{r} 1.55\ldots \\ 9\overline{)\,14} \\ \underline{-9} \\ 50 \\ \underline{-45} \\ 50 \\ \underline{-45} \\ 5 \end{array}$$

$\therefore 1\frac{5}{9} = 1.555\ldots = 1.\dot{5}$

$\Big\}$ $02\frac{1}{2}$ marks

8. (a) Absolute value of $\frac{1}{4} + \frac{1}{2} - 2$

$= \left|\frac{1}{4} + \frac{1}{2} - 2\right|$

$= \left|\frac{1+2}{4} - 2\right|$

$\Big\}$ 01 mark

(8)(a)

$= \left|\frac{3}{4} - \frac{2}{1}\right|$ ------ (01 mark)

$= \left|\frac{3-8}{4}\right|$ ------ (01 mark)

$= \left|\frac{-5}{4}\right|$ ------ (01 mark)

$= \frac{5}{4}$ ------ (01 mark)

∴ The absolute value of $\frac{1}{4} + \frac{1}{2} - 2$ is $\frac{5}{4}$

(b) $|3-5| + |12 \div 3| - 5 \times 2$

$= |-2| + |4| - 5 \times 2$ ------ 02 marks

$= 2 + 4 - 10$ ------ 01 marks

$= 6 - 10$ ------ 01 mark

$= \underline{-4}$ ------ 01 mark

9 (a) Monthly salary = 270,000

food $= \frac{2}{3} \times 270,000 = 180000$ ---- (01 mark)

pair of shoes $= \frac{1}{2}(270,000 - 180000)$ ---- (02 marks)

$= \frac{1}{2} \times 90,000$

$= 45000$ -------- 01 mark

∴ Tsh. 45000 was used to buy pair of shoes. - - - - - - (01 mark)

(b) Fraction of salary used to buy food =

$= \frac{45,000}{270,000}$ (02 mark)

$= \frac{1}{6}$

```
      0.166...
 6 ⟌ 1
    -0
     10
    - 6
      40
     -36
       40
      -36
        4
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(02 marks)

(5) Fraction in decimal is $0{\cdot}1\dot{6}$ — ~~[illegible]~~ (01 mark)

0·

Number of bottles = 12
Cost of one bottle = 600 shillings
Total cost = 12 × 600
= 7200 shillings } 03 marks

Number of people = (Amina + 3 friends)
= 1 + 3
= 4. } 02 marks

Amount each person paid

$$= \frac{7200 \text{ shillings}}{4}$$ --- 03 marks

= 1800 shillings --- 01 mark

∴ Each person paid 1800 shillings - 01 mar